

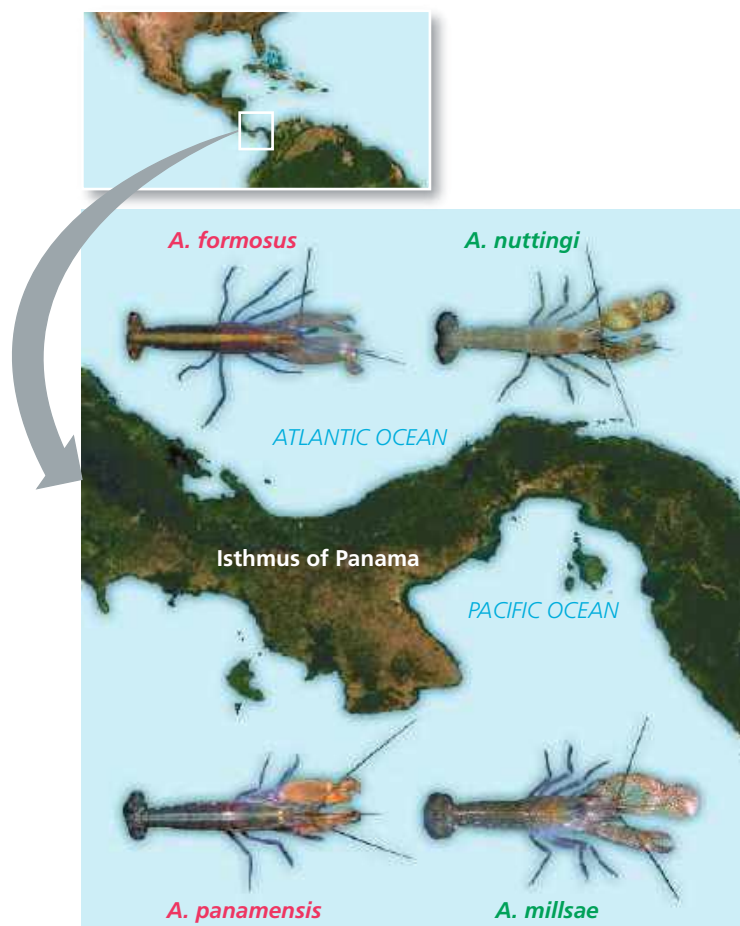
to reproduction between mosquitofish from ponds with predators and those from ponds without predators. Thus, as a by-product of selection for avoiding predators, reproductive barriers have formed in these allopatric populations.

### Evidence of Allopatric Speciation

Many studies provide evidence that speciation can occur in allopatric populations. For example, laboratory studies show that reproductive barriers can develop when populations are isolated experimentally and subjected to different environmental conditions (Figure 24.7).

Field studies indicate that allopatric speciation also can occur in nature. Consider the 30 species of snapping shrimp in the genus *Alpheus* that live off the Isthmus of Panama, the land bridge that connects South and North America (Figure 24.8). Fifteen of these species live on the Atlantic side of the isthmus, while the other 15 live on the Pacific side. Before the isthmus formed, gene flow could occur between the Atlantic and Pacific populations of snapping shrimp. Did the species on different sides of the isthmus originate by allopatric speciation? Morphological and genetic data group these shrimp into 15 pairs of *sister species*, pairs whose member species are each other's closest relative. In each of these

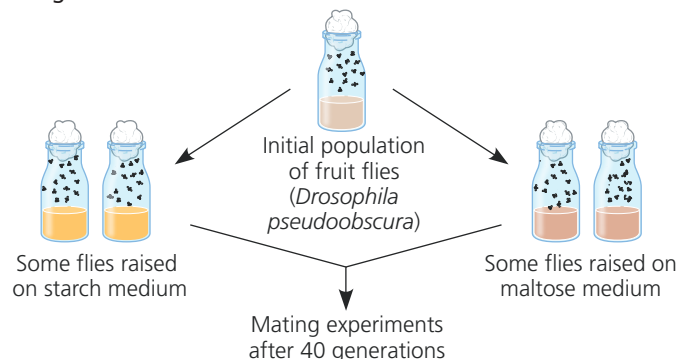
▼ **Figure 24.8 Allopatric speciation in snapping shrimp (*Alpheus*).** The shrimp pictured are just 2 of the 15 pairs of sister species that arose as populations were divided by the formation of the Isthmus of Panama. The color-coded type indicates the sister species.



### ▼ Figure 24.7 Inquiry

#### Can divergence of allopatric populations lead to reproductive isolation?

**Experiment** A researcher divided a laboratory population of the fruit fly *Drosophila pseudoobscura*, raising some flies on a starch medium and others on a maltose medium. After one year (about 40 generations), natural selection resulted in divergent evolution: Populations raised on starch digested starch more efficiently, while those raised on maltose digested maltose more efficiently. The researcher then put flies from the same or different populations in mating cages and measured mating frequencies. All flies used in the mating preference tests were reared for one generation on a standard cornmeal medium.



**Results** Mating patterns among populations of flies raised on different media are shown below. When flies from "starch populations" were mixed with flies from "maltose populations," the flies tended to mate with like partners (see the lefthand table). But in a control group consisting of flies from two different populations adapted to starch, the flies were about as likely to mate with flies of the other population as with flies of their own population (see the righthand table); similar results were obtained for control groups adapted to maltose.

|      |         | Female |         |
|------|---------|--------|---------|
|      |         | Starch | Maltose |
| Male | Starch  | 22     | 9       |
|      | Maltose | 8      | 20      |

Number of matings  
in experimental group

|      |                     | Female              |                     |
|------|---------------------|---------------------|---------------------|
|      |                     | Starch population 1 | Starch population 2 |
| Male | Starch population 1 | 18                  | 15                  |
|      | Starch population 2 | 12                  | 15                  |

Number of matings  
in control group

**Conclusion** In the experimental group, the strong preference of "starch flies" and "maltose flies" to mate with like-adapted flies indicates that a reproductive barrier was forming between these fly populations. Although this barrier was not absolute (some mating between starch flies and maltose flies did occur), after 40 generations reproductive isolation appeared to be increasing. This barrier may have been caused by differences in courtship behavior that arose as an incidental by-product of differing selective pressures as these allopatric populations adapted to different sources of food.

**Data from** D. M. B. Dodd, Reproductive isolation as a consequence of adaptive divergence in *Drosophila pseudoobscura*, *Evolution* 43:1308–1311 (1989).

**WHAT IF?** Why were all flies used in the mating preference tests reared on a standard medium (rather than on starch or maltose)?